Limitations & Caveats of the Study

1. Satellite Measurement Bias (WorldPop)

Caveat	WorldPop underestimates population in dense urban areas due to vertical density blindness — satellites capture building footprints, not floors or families within.
Example	In Mubarak Pur Dabas, WorldPop estimates ~6,300 children vs. ~17,900 enrolled students in UDISE+.
Implication	Distorts model results in high-density wards; likely underestimates true demand.

2. Private School Blind Spot

Caveat	Only government schools are modelled; private schools are excluded.
Example	Wards with many private schools may appear underutilized in the government system.
Mitigation	Zone dummy variables are used to absorb regional private-school effects (e.g., South Delhi).
Implication	Demand may be underestimated in areas with high private school presence.

3. 40-Seat Capacity Assumption (RTE Norm)

Caveat	Seat capacity is computed as Total Classrooms × 40, per RTE guidelines — an upper limit that may exceed practical capacity.
Example	A 10-classroom school is assumed to hold 400 students, even if realistic capacity is lower.
Implication	Overcrowding may be underestimated; results are conservative.

4. Omission of Senior Secondary Schools

Caveat	74 schools excluded for lacking ward identifiers, **40.5% (30schools)** were **Senior Secondary (Classes 9–12)** institutions. This indicates a structural bias where UDISE+ data is less reliable for high-school level mapping compared to primary levels.
Example	These schools could not be spatially assigned to specific wards.
Implication	Findings are more reliable for primary and middle levels; capacity may be slightly underestimated for higher grades.

5. Map Boundary Errors (Sliver Gap Problem)

Caveat	Small digitization errors in the ward boundary GeoJSON create sliver gaps between adjacent wards, breaking spatial connectivity.
Example	Ward 9 (Gharoli) appeared as an isolated island at a 150 m threshold.
Mitigation	300 m fuzzy contiguity weights matrix applied to bridge gaps.

Implication	A 300 m fuzzy contiguity matrix was the minimal threshold needed to restore a fully connected weights matrix without artificially linking non-adjacent wards.

6. Infrastructure Adequacy vs. Service Delivery

Caveat	The Mismatch Index (MI) measures physical capacity (classrooms), not educational quality.
Example	A ward with MI = 1.0 may still lack toilets, libraries, or science labs.
Implication	Analysis captures infrastructure adequacy only — not learning outcomes or service quality.

7. Temporal Data Gap (4-Year Lag)

Caveat	School data is from 2023–24; WorldPop population data is from 2020.
Example	Fast-growing wards like Rohini or Bawana may have seen significant population change in the interim.
Implication	The population variable is lagged, reducing accuracy in high-growth areas.

8. Exclusion of Zero-School Wards

Caveat	9 Wards with no government schools are excluded from regression (MI is undefined for them).
Example	Ward No.120 has ~40,931 children but zero government schools.
Implication	The most extreme mismatches remain invisible in the model, leading to underestimation of the overall crisis.

9. Modifiable Areal Unit Problem (MAUP)

Caveat	Results are sensitive to the choice of spatial unit; switching from wards to districts or grid cells can alter Moran's I and regression coefficients.
Implication	Findings are specific to the ward level and may not generalise to other spatial scales.

10. The "Chicken-and-Egg" Problem (Reverse Causality)

Caveat	We found that teacher shortages and overcrowding happen together, but we don't know which one **started** first.
Implication	It's a loop: A lack of teachers makes a school overcrowded, and that overcrowding makes teachers stressed and want to leave. Our model shows they are linked, but it can't prove who is the original "cause."